Dashboard UI & Feature Breakdown

This technical guide details the interface components, interactive views, and specific functions of the **Climate-Resilient Yield Prediction Platform (CR-YPP)** SaaS dashboard.

1. Field Selection & Region Controls

Region & Crop Parameter Selector

Allows users to switch between agricultural test zones (e.g., Greater Noida Zone, Yamuna Expressway Sector) and target crop varieties (Wheat during Rabi season, Paddy during Kharif season) to initialize custom regional model inference runs.

2. Geospatial Map & Risk Heatmap View

Interactive Map Viewport & Grad-CAM Overlays

Displays multi-spectral Sentinel-2 satellite imagery overlaid with color-coded risk gradients. High-yield zones appear in dark green, while stress-vulnerable zones appear in amber and red. Users can toggle Explainable AI (XAI) Grad-CAM overlays to inspect spatial attention maps.



3. Yield Prediction & Confidence Summary

Harvest Metric Cards

Instantly reports computed harvest metrics in Tons per Hectare (T/ha), accompanied by model confidence percentages (e.g., 92%) derived from the XGBoost multimodal late-fusion ensemble.

4. Meteorological & Climate Stress Analytics

Time-Series Anomaly Charts

Visualizes 30-day NASA meteorological trends, tracking temperature anomalies (e.g., +1.4°C) and precipitation stress alongside historical resilience scores across consecutive seasons.



5. Action Toolbar & Automated Alerting

Interactive Tools & Alerts

Provides quick-action buttons for drawing custom field boundary polygons, toggling raster data layers, generating downloadable PDF reports, and triggering automated severe weather alerts.